



सत्यमेव जयते

**Government of India**  
**Ministry of Statistics & Programme Implementation**  
**National Statistics Office**

**Report on**  
**Model-Based District-Level Estimates Based on the**  
**Household Consumption Expenditure Survey (HCES):**  
**2022-23**  
**For Uttar Pradesh**

**National Statistics Office**  
**Ministry of Statistics and Programme Implementation**

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## **EXECUTIVE SUMMARY**

The Household Consumption Expenditure Survey (HCES) is a renowned statistical initiative taken by the Government of India which aims to give comprehensive information about the consumption habits of households throughout the nation. This research uses the 2022–23 HCES to estimate MPCE at the district level for Uttar Pradesh. Traditionally used for estimating state- and national-level MPCE and budget shares, the 2022–23 HCES has been leveraged in this report to estimate district-level MPCE for Uttar Pradesh. The main goal is to use model-based techniques, specifically Small Area Estimation (SAE), to fill the data gap at the district level. Decentralized planning, welfare targeting, and evidence-based governance all depend on the district-level estimates.

The survey covered 1,096 villages and 598 urban blocks in Uttar Pradesh, totaling 30,238 households (rural: 19,611; urban: 10,627). A stratified multi-stage design was employed using Simple Random Sampling Without Replacement (SRSWOR). While the sample size was adequate for state-level estimates, it was insufficient for district-level direct estimates. Therefore, the report adopts the Fay–Herriot (FH) and Spatial Fay–Herriot (SFH) models under the SAE framework to borrow strength from covariate data and neighboring districts to generate reliable small area statistics.

Several administrative and scheme-based indicators were used as covariates, including:

- Rural areas: Number of old age pension beneficiaries, Ayushman Bharat (PM-JAY) patient counts.
- Urban areas: Number of LPG connections, Antyodaya beneficiaries and food distribution.
- Spatial effects: Contiguity of districts used to model influence between neighboring areas in SFH models.

All covariates were taken for the reference year 2022–23.

Different statistical models are applied on the data to get the more reliable results.

1. **Direct estimates** were computed from HCES but were unreliable due to small sample sizes.
2. **Fay–Herriot (FH)** model: Used area-level auxiliary variables to improve precision.

3. **Spatial Fay–Herriot (SFH)** model: Added spatial correlation between neighboring districts using a **Simultaneous Autoregressive (SAR)** process.
4. Variance estimates were generated using **bootstrap techniques** due to the unavailability of detailed sampling weights.

Both FH and SFH models significantly reduced the Coefficient of Variation (%CV), improving reliability over direct estimates. The SFH model showed even greater improvement in % CV, especially in rural areas, by incorporating neighborhood effects. Estimates could be produced for areas like Shrawasti Urban, which had no sampled units, using model-based predictions. Top 5 Districts by MPCE (rural) are Bagpat, Saharanpur, Gb Nagar, Meerut, Ghaziabad. Top Districts by MPCE (urban) are Gb Nagar, Gonda, Ghaziabad, Bagpat, Lucknow.

Model-based estimation for smaller domains can be a powerful tool for taking data-driven and informed policy decisions across various domains. However, the limitations and potential biases of models and consequences arising out of these are required to be acknowledged and considered to ensure that model-based estimation serves the intended purpose in policymaking:

- Model accuracy is highly contingent on the quality and relevance of covariates used.
- Estimates are sensitive to model assumptions, and misspecification can lead to misleading results.
- Users must be clearly informed about the nature of these estimates, including uncertainty measures, and distinctions from design-based statistics.

SAE techniques make district-level socio-economic planning feasible without using large sample sizes. Model-based estimates are useful for targeting subsidies, evaluating regional disparities, and monitoring SDG indicators. These estimates enable data-driven decision-making at sub-state levels (e.g., blocks, villages). This study demonstrates the power of model-based estimation, particularly SAE, in delivering statistically sound and granular economic indicators like MPCE at the district level. The approach is scalable and cost-effective, ideal for India's push toward localized planning. SAE offers a path toward data equity, enabling better governance by ensuring that no district is statistically invisible.

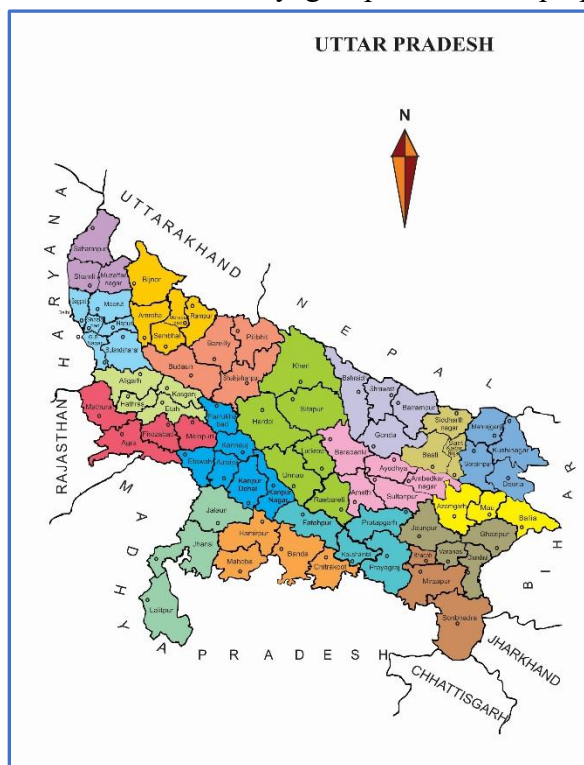
# **CHAPTER 1**

## **INTRODUCTION**

# 1 INTRODUCTION

## 1.1 Background

The **Household Consumption Expenditure Survey (HCES)** is globally renowned for providing information of household consumption (goods as well as services). This generic survey is also rich source of information on household characteristics and demographic particulars of the members. The Information thus collected is actively used in understanding the consumption patterns, budget shares of commodity groups, used for preparation of the weighting diagram for compilation



Consumer Price Indices (CPIs). The data collected in HCES is also utilized, inter alia, for deriving various other macroeconomic indicators. Technically the survey is so acknowledged by Deaton & Kozel, 2005<sup>1</sup> “These surveys, pioneered by P. C. Mahalanobis in the 1940s and 1950s (Mahalanobis and Sen 1954), were the world’s first system of household surveys to apply the principles of random sampling established in the 1920s and 1930s.”

To facilitate a decision on the choice of an appropriate period for the revision of the ‘base year’ for macroeconomic indicators, the National Statistical Commission (NSC) recommended to undertake HCES during July, 2020 to June, 2021. and 2021-22. However, due to the emergence of Covid-19 pandemic, the survey of 2020-21 could not be started. The first of the two consecutive surveys have been planned during the period

August, 2022 to July, 2023 and August, 2023 to July, 2024. The survey has been renamed as “Household Consumption Expenditure Survey (HCES)” from erstwhile “Household Consumer Expenditure Survey” as suggested by the NSC.

The HCES survey is designed to give all-India and state-wise MPCE estimates. To obtain the sub-state level, especially district level estimates. it was recommended that MoSPI to explore the possibility to support the States.

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<sup>1</sup> Deaton, Angus, and Valerie Kozel. 2005. “Data and Dogma: The Great Indian Poverty Debate”. Washington, D.C.: World Bank.

## 1.2 Formation of Committee

In pursuance of the recommendation of the National Statistical Commission (NSC), in its 129th Meeting held on 30th October, 2023, a Steering Committee for NSS has been constituted under the Chairmanship of NSC Chairman with tenure 2 years. Further, the Terms of Reference (ToR) of the Committee is furnished below:

### **Make recommendations to NSC in respect of the following:**

- a. To review the survey methodology including questionnaire, sampling frame, sampling design, concepts and definitions, survey instruments and recommend changes if needed on these and related questions.
- b. To examine the issues or any other matters raised from time to time on the subject methodology, etc. related to all National Sample Surveys by MOSPI or external entities as brought before the Steering Committee.
- c. To suggest conduct of pilot surveys/ pre-testing of survey instruments, if necessary, before finalizing from time to time.
- d. To undertake study to identify data gaps/ additional data requirements, if any, in respect of surveys and suggest appropriate strategy for improvement.
- e. To provide technical guidance to the Central and State (wherever State participation is there) level agencies for the conduct of the surveys.

2. In the second meeting of the Steering Committee for NSS survey held on 25.07.2025, the committee had recommended that NSSO, MoSPI may undertake a pilot study on the feasibility for generating the model-based district-level estimates based on the recent HCES 2022-23 for the State of U.P. **In this regard, a committee was formed under the Chairmanship of the Dr. Mausumi Bose, former Professor, Indian Statistical Institute (ISI) Kolkata with members from the NSSO and DES, UP.**

## 1.3 The Terms of Reference of the committee

The committee had been tasked to deliberate various issues related to the generation of the Model-based district-level estimates of MPCE. The committee had to conduct a pilot study for exploring the feasibility of generation of Model-based district-level estimates based on the recently released data of the HCES 2022-23 for the State of U.P. The terms of reference (ToR) of the committee were to undertake the technical and analytical work related to the generating Model-based district-level estimates.

## 1.4 Members of Committee

The committee has the following composition:

| S. No. | Committee Members Name, Designation and Department/Ministry/Institution: | Chairman/Members        |
|--------|--------------------------------------------------------------------------|-------------------------|
| 1.     | Prof. Mausumi Bose, former Professor, ISI Kolkata.                       | Chairman (non-official) |
| 2.     | Prof. Bornali Bhandari, NCAER, Delhi.                                    | Non-official member     |

|    |                                                                                                          |                             |
|----|----------------------------------------------------------------------------------------------------------|-----------------------------|
| 3. | Dr. Laxmi Kant Dwivedi, Professor & Head, International Institute for Population Science (IIPS), Mumbai. | Official Member             |
| 4. | Prof. Vikram Goyal, IIIT Delhi.                                                                          | Official Member             |
| 5. | Shri Salil K. Mukhopadhyay, ADG, Enterprise Survey Division, NSSO, MoSPI.                                | Official Member             |
| 6. | Dr. Amitava Saha, DDG, Household survey Division, NSSO, MoSPI.                                           | Official Member             |
| 7. | Dr. K.V. Raju, Economic Advisor to Chief Minister, Uttar Pradesh.                                        | Official Member             |
| 8. | Mrs. Alka Bahuguna Dhoundiyal, Additional Director, Economics and Statistics Division, Uttar Pradesh.    | Official Member             |
| 9. | Shri. M. S. Subramanya Rao, DDG, C&QCD, NSSO, MoSPI.                                                     | Member-Secretary (Official) |

Further a separate technical team, comprising ISS officers and officers from DES, UP had been formed on 17.03.2025 to undertake the technical and analytical work related to the committee.

The team has the following composition:

| S. No. | Team Member Name         | Designation and Department/Ministry          |
|--------|--------------------------|----------------------------------------------|
| 1.     | Shri Sujeet Pujari       | Joint Director, EnSD, NSO, MoSPI             |
| 2.     | Dr. Priyanka Anjoy       | Deputy Director, NAD, NSO, MoSPI             |
| 3.     | Ms. Priya Devwanshi      | Deputy Director, HSD, NSO, MoSPI             |
| 4.     | Smt. Manju Ashok         | Deputy Director, DES-UP, HQ                  |
| 5.     | Dr. Rajesh Kumar Chauhan | Economics and Statistics Officer, DES-UP, HQ |
| 6.     | Ms. Yashi Pal            | Economics and Statistics Officer, DES-UP, HQ |

## 1.5 Objective of the Report

The primary objective of this report is to develop district-level estimates of Monthly Per Capita Consumption Expenditure (MPCE) for all districts of Uttar Pradesh using data from the HCES 2022–23 survey and other auxiliary covariates.

- To generate district-level estimates of Monthly Per Capita Consumption Expenditure (MPCE) for all districts of Uttar Pradesh using HCES 2022-23 survey data.
- To integrate auxiliary covariates to enhance the accuracy and granularity of MPCE estimates at the district level.



- To assess the feasibility of using model-based estimation approaches for generating district-level socio-economic estimates in Uttar Pradesh.
- To provide actionable insights for policy planning, welfare schemes, and regional development initiatives based on district-level MPCE estimates.

## 1.6 HCES 2022-23 SURVEY COVERAGE

### Geographical Coverage

At all India level the survey was conducted in 8,723 villages and 6,115 urban blocks covering 2,61,746 Households across India. Of these, 1,55,014 households were in rural areas and 1,06,732 in urban areas. Among above mentioned villages and urban blocks, Uttar Pradesh covered 1,096 villages and 598 urban blocks. In Uttar Pradesh data was collected from a total of 30,238 households, with 19,611 in rural areas and 10,627 in urban areas.

It is to be noted that HCES 2022-23 was designed on basis of the census 2011 (at that time UP had 71 districts). At present state's current administrative structure includes 75 districts. This figure includes the four districts- Sambhal, Shamli, Hapur, and Amethi - which were newly created by carving out areas from existing districts, primarily around the 2010-2012 period. **Sambhal** was carved from Moradabad and Budaun districts, **Shamli** was carved from Muzaffarnagar district and **Hapur** was carved from Ghaziabad district (all these three new districts were created on 28 September 2011). District **Amethi** was carved from Sultanpur and Raebareli districts (created on 1 July 2010).

### Population Coverage

The survey primarily covered all resident households across India.

## 1.7 QUESTIONNAIRES FOR THE SURVEY

The Household Consumption Expenditure Survey (HCES): 2022-23 had four distinct questionnaires for data collection:

- FDQ (Food items)
- CSQ (Consumables and Services items)
- DGQ (Durable goods)
- HCQ (Household Characteristics and Demographic particulars)

The FDQ, CSQ, and DGQ, covering the three main consumption categories, were canvassed in selected households across three separate, consecutive monthly visits. All possible six sequencing of the three questionnaires, namely, [(Q1, Q2, Q3), (Q1, Q3, Q2), (Q2, Q1, Q3), (Q2, Q3, Q1), (Q3, Q1, Q2) and (Q3, Q2, Q1)], where Q1 refers to FDQ, Q2 refers to CSQ and Q3 refers to DGQ, were used in the survey to eliminate any bias due to adoption of any particular sequencing of the questionnaires.

## 1.8 Covariates under study

Towards this, the committee considered generation of MPCE estimates for each district in UP, separately for the rural and urban parts in each 75 districts, using the published HCES 22-23 survey data and data on several covariates which were given by DES, UP. Following administrative data related to household characteristics that influence consumption of household from various schemes were collected and studied.

| S. No. | Covariates                                                                                                                                             |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.     | Electricity consumption Indicator- Number of Commercial connections (Rural/Urban)                                                                      |
| 2.     | Public Distribution System - Number of HH benefited under Domestic/ Commercial LPG connection (Rural/Urban)                                            |
| 3.     | Domestic LPG Gas Consumption (in MT) (Rural/Urban)                                                                                                     |
| 4.     | Transport Department. - Number of Domestic/Commercial CNG connection                                                                                   |
| 5.     | Public Distribution System - Number of households (HH) benefited under NFSA(Rural/Urban),<br>No. of Ration card and no. of beneficiaries (Rural/Urban) |
| 6.     | Number of Active SHGs (Rural/Urban)                                                                                                                    |
| 7.     | MNREGA: Mahatma Gandhi National Rural Employment Guarantee Act (Only Rural)                                                                            |
| 8.     | Old age Pension (Rural/Urban)                                                                                                                          |
| 9.     | Widow Pension (Rural/Urban)                                                                                                                            |
| 10.    | Divyangjan Pension (Rural/Urban)                                                                                                                       |
| 11.    | DBT: Direct Beneficiaries Transfer- PM Kisan Samman Nidhi (Rural)                                                                                      |
| 12.    | PM Awas Yojana (Rural/Urban)                                                                                                                           |
| 13.    | Number of Ayushman Bharat card Beneficiaries and amount disbursed (Rural/Urban)                                                                        |
| 14.    | District Domestic Product (DDP) 2022-23                                                                                                                |
| 15.    | List of divisions and their belonging districts of Uttar Pradesh                                                                                       |
| 16.    | Colored Map of Uttar Pradesh                                                                                                                           |
| 17.    | Number of Employees in Local Bodies and cantonment board (Urban)                                                                                       |
| 18.    | Number of Private Basic/Primary Schools and the number of students (Rural/Urban)                                                                       |
| 19.    | Number of Private Madhyamik /Secondary/Higher Secondary Schools and the number of students (Rural/Urban)                                               |

| S. No. | Covariates                                                                      |
|--------|---------------------------------------------------------------------------------|
| 20.    | Number of Private Engineering Colleges and the number of students (Rural/Urban) |

Further correlation was computed between covariates and MPCE to see relevant correlation with the MPCE. Only those covariates which showed significant correlation with MPCE were used for study.

### Covariates which were used by committee

#### 1. Rural Areas-

- a. Old age pension (Samaj Kalyan Vibhag): Number of Beneficiaries-
- b. Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB PM-JAY)-  
Number of patients

#### 2. Urban Areas-

- a. **Domestic LPG Gas Connection:** Number of households using Domestic LPG connection.
- b. **Antyodaya:** Number of Beneficiaries and food distributed (in Ton) to Antyodaya card holders.
- c. **Spatial Covariates-** For 'borrowing strength' from neighboring areas, districts with common boundaries, distance between districts, districts within same Mandals were considered and finally districts with common boundaries were used as 'neighbours'.

## 1.9 REFERENCE PERIOD OF THE COVARIATES USED

The reference periods used in this study for different covariate was financial year 2022-23 (April 22-March 23).

### 1.10 Monthly per capita consumption expenditure (MPCE)

The Monthly Per Capita Consumption Expenditure (MPCE) is calculated by dividing a household's total monthly consumption expenditure by its size, this suffices measuring standard of living. For individuals, their MPCE is simply that of their household, meaning the survey overlooks intra-household consumption.

### 1.11 Illustration of Covariates

#### For Rural Areas-

- A. **Old age pension (Samaj Kalyan Vibhag): Number of Beneficiaries-** Government transfers the pension amount Rs. 1000 to the beneficiaries in their respective bank accounts.

Eligibility: a. Age should be 60 years or more

b. Applicant should be below poverty line or his/her annual income should be

up to Rs. 46,080 in rural areas and Rs. 56,460 in urban areas.

**B. Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB PM-JAY)-**

Ayushman Bharat Pradhan Mantri Jan Arogya Yojana is the world's largest health insurance scheme. It provides health coverage of up to ₹5 lakh per family annually to access quality healthcare in case of hospitalization.

Eligibility: To qualify for the Ayushman card under the Ayushman Bharat Yojana, one must belong to one of the specified categories: SC/ST, lower income group or EWS. Additionally, all senior citizens aged 70 years or more, regardless of income or asset requirements are eligible for the Ayushman card.

**For Urban Areas-**

- A. Domestic LPG Gas Connection:** Number of household benefited under Domestic/ Commercial LPG connection.
- B. Antyodaya:** Allocated to the poorest families with no stable source of income. Under this scheme, the families below BPL are given 35 kg of food grains each month.

# **CHAPTER 2**

## **Significance of the Study for India**

## 2 Why is the Research Study Important for India?

### Enhancing District-Level Data Accuracy through Small Area Estimation (SAE)

Decentralized planning, evidence based policymaking, and efficient monitoring of development initiatives in India require precise and comprehensive socio-economic data at the sub-state level, especially district and sub-district levels. While estimates at the national and state levels are often produced using administrative records and extensive surveys, estimates at the district level are frequently constrained by a lack of comprehensive data, inconsistent reporting, or a small sample size.

In scenarios where standard direct estimates are statistically erroneous or unavailable, a class of statistical approaches known as **Small Area Estimation (SAE)** was developed to produce precise estimates for small geographic or demographic subgroups (also known as "small areas"). Auxiliary characteristics, frequently from administrative or census sources, are combined with restricted survey data in SAE to generate more accurate and reliable estimates at the required degree of disaggregation.

### 2.1 Small Area Estimation (SAE) to Measure Economic Indicators at Local Levels

SAE is a robust statistical technique that let researchers and policymakers to get precise projections for local-level socio-economic indicators. SAE utilizes auxiliary data, or factors that are connected with the target indicator, to enhance the estimating process rather than depending solely on direct survey measurements. This enables the micro scale estimation of a variety of economic indices, including employment levels, poverty rates, monthly per capita expenditure (MPCE), gross value added (GVA), and district domestic product (DDP).

Integration of several data sources, including administrative records, census records, remote sensing inputs, and survey-based indicators, is one of the major benefits of SAE. For instance, auxiliary indicators that may be used for assessing DDP or GVA at the district level include:

- a. The quantity of businesses listed in the Economic Census
- b. Statistics on agricultural output derived from crop surveys;
- c. Electricity usage, which represents business and industrial activities
- d. MSME funding and credit flow, which show company expansion, and many more.

In the same way, SAE models can use inputs like housing settings, education levels, household asset ownership, or local price indices to estimate consumption-related metrics like MPCE. Despite being indirect, these proxies contribute in improving estimates in situations when local household consumption surveys may not be sufficient.

The use of SAE-based estimations are extensively used where direct data are unavailable, SAE makes it possible to create lower level (e.g. district-level) socio-economic profiles that guide local government in issues of paramount interest like fiscal capacity evaluations, formulating regional development plans, and making public investment choices. In modern day's context SAE-based MPCE or poverty estimations are essential for determining the scope of social welfare initiatives, in rendering subsidies, and quantifying inequalities at the micro level. The granularity required to measure progress in rural or underserved regions is another way that these estimates aid in the monitoring important indices like Sustainable Development Goals (SDGs).

In summary, Small Area Estimation provides a statistically robust and policy-relevant solution to the data gaps that exist in regional economic measurement. By integrating multiple data sources and applying advanced modeling techniques, SAE helps generate credible estimates for key socio-economic indicators-ranging from DDP and GVA to MPCE and poverty-across small areas where traditional data collection is limited or cost ineffective. This not only strengthens evidence-based policymaking but also ensures that development efforts are suitably targeted and thus efficiently implemented.

## 2.2 SAE in MPCE Estimation Using HCES 2022–2023: A Recent Case Study

The use of SAE in India's socio-economic planning framework is demonstrated by a recent pilot study. A technical committee headed by **Dr. Mausumi Bose**, former Professor, Indian Statistical Institute (ISI) Kolkata and representatives from the **Directorate of Economics and Statistics (DES), UP**, **Dr. K.V. Raju** (Economic Advisor to the Hon'ble Chief Minister of Uttar Pradesh), and **NSO, MoSPI** made the recommendations that guided the study.

The purpose of this study is to investigate model-based ways to derive district-level estimates using inputs from the Household Consumption Expenditure Survey (HCES) 2022-23. A pilot study was undertaken to produce Monthly Per Capita Consumption Expenditure (MPCE) estimates for all districts in Uttar Pradesh, disaggregated by rural and urban areas, in tune to the committee's recommendations. This was performed by utilizing SAE techniques that integrated district-level

covariates provided by DES, UP, as well as state-level HCES survey data. The HCES survey sample size was statistically sufficient for the state-level estimation but insufficient for the district-level estimation, and so direct district-level MPCE estimates were unreliable. This presented a challenge to generate reliable estimates at district level. In order to generate reliable district-level estimates, the SAE technique successfully addressed this problem by combining auxiliary variables with the available survey data. The increasing need for evidence-based district-level planning and governance was supported by this methodology, which made sure that localized estimates were both statistically sound and pertinent to policy.

Small Area Estimation has rapidly emerged as a transformative methodology in the Indian statistical and policy ecosystem. From estimating **DDP and MPCE** to addressing limitations in **employment and enterprise surveys**, SAE enables the transition from **macro-level insights** to **micro-level action**. As India advances toward digital governance, real-time planning, and inclusive growth, SAE will be indispensable in equipping decision-makers with **localized, actionable, and statistically robust data supply**.

Investing in SAE capacity- through training, computational tools, and methodological innovations will be vital for strengthening India's decentralized development modelling and data-driven governance.

With the aid of this research study, district level estimates can be derived from various socio-economic surveys where State level estimates are available. Thus, it will fulfil the need of lower level estimation i.e. district/block/tehsil/panchayat for achieving the objective of good governance.



# **CHAPTER 3**

# **INTERNATIONAL**

# **PRACTICES**

## 3 International Practices for generating model-based district-level estimates

Generating model-based district-level estimates is a common practice in official statistics, especially when sample surveys are not large enough to produce reliable direct estimates at lower administrative levels. This is particularly relevant for sectors like health, education, poverty, employment, agriculture, and demographic indicators. Below are the key international practices followed by statistical agencies and researchers:

### 3.1 Use of Small Area Estimation (SAE) Methods

Most countries and organizations use Small Area Estimation methods when sample sizes are small for subnational areas like districts. These are model-based approaches that "borrow strength" from related areas or auxiliary information.

#### ➤ **Model-Based SAE**

- **Fay-Herriot Model (Area-Level SAE):** Common when auxiliary data is available at the district level.
- **Empirical Best Linear Unbiased Prediction (EBLUP):** A widely used approach within the Fay-Herriot framework.
- **Bayesian Hierarchical Models:** Increasingly popular due to their flexibility and ability to provide uncertainty estimates.

#### ➤ **Unit-level model**

- **Battese-Harter-Fuller Model:** Used when unit-level data (e.g., household survey data) is available along with covariates.
- Requires detailed survey microdata and matching auxiliary data.

#### **1. Integration of Survey and Census/Administrative Data**

- Survey data (e.g., DHS, MICS, labor force surveys) often lack sufficient sample size at the district level.
- Census or administrative records are used as auxiliary variables in the modeling.

These may include:

- Satellite imagery (e.g., night-time lights)
- GIS-derived population density
- School or health facility coverage
- Tax or welfare registration data

Example: The World Bank's "Poverty Mapping" combines household survey data with census data using SAE.

## 3.2 Use of Geospatial Modeling

- Popular in health and nutrition estimates (e.g., malaria prevalence, child malnutrition).
- Bayesian geostatistical models allow predictions over space, accounting for spatial correlation.
- Tools like INLA (Integrated Nested Laplace Approximation) and R-INLA, Bayesian Additive Regression Trees (BART), or Gaussian processes are used.

Example:

IHME (Institute for Health Metrics and Evaluation) produces district-level health estimates globally using geospatial modeling.

GADM and WorldPop datasets are often used for spatial boundaries and population weights.

## 3.3 International Organizations and Best Practices

### ➤ **World Bank:**

- Uses SAE for poverty mapping (e.g., in collaboration with national statistical offices).
- Toolkit: PovMap, SAE in R (e.g., 'emdi', 'sae' packages)

### ➤ **UNICEF & UNFPA:**

- Use Bayesian SAE for child and maternal health indicators.
- Often use MICS or DHS survey data combined with census covariates.

### ➤ **Organisation for Economic Co-operation and Development (OECD) and Eurostat:**

- Produce regional and sub-regional statistics using model-based estimations.
- Eurostat's **SAE Toolbox** is widely used in Europe.

# **CHAPTER 4**

# **METHODOLOGY**

## 4 Methodology

### 4.1 MPCE Estimation Approach

The major objective of the present study is to generate model-based district-level estimates of MPCE for the state of Uttar Pradesh using appropriate Small Area Estimation (SAE) approach. It is a reasonably good supposition that, MPCE are not uniformly distributed across the regions, i.e., in districts cross classified by Rural and Urban sectors. The traditional survey estimation approaches are not suitable for obtaining the reliable estimates at this level because of sample size problem. It is the main endeavor of Small Area Estimation (SAE) approach to produce micro level statistics with considerably acceptable precision without incurring any extra cost and utilizing existing survey data. The SAE approach is particularly known for its ability to produce efficient and reliable small area level estimates.

For this study, the HCES 2022-23 survey of National Statistics Office (NSO), Ministry of Statistics and Programme Implementation (MOSPI) has been used. In India, HCES survey is the main and detailed source of official statistics on household level monthly and item-wise consumption expenditure and this survey is planned to produce statistics at State and National level. The sampling design used in the HCES survey is stratified multi-stage random sampling considering villages/urban blocks or sub-units of these as the First Stage Units (FSU) and the households as the Ultimate Stage Units (USU). Simple Random Sampling Without Replacement (SRSWOR) method is used for selecting both the FSUs and households. In order to ensure proper representation of households of different economic categories, all the households of a selected village/urban block are classified into three groups depending on a criterion based on (i) land possessed in rural areas and (ii) possession of car in urban areas as on the date of the survey. A total of 18 households with proportional representation from the three groups were selected in HCES 2022-23 survey. The detailed survey Methodology as well as Estimation procedure of HCES 2022-23 is available in the website of MOSPI (<https://microdata.gov.in/NADA/index.php/catalog/224>) and specifically microdata section comprises the unit level data of this survey and other relevant documents.

The Monthly Per Capita Consumption Expenditure (MPCE) data from HCES survey broadly captures the living standard of a household through enlisting their monthly consumption on various food and non-food items. However, through this survey, efficient estimates of MPCE and other

related parameters can be obtained at State and National level for both rural and urban sectors only. The reason is, with the existing sampling design and sample sizes district level direct estimates will be unstable, since within each district sample size is not large enough to produce district level estimates with adequate precision and reliability. A large overall sample size is not going to be a feasible approach, because that may leave other domains with small sample sizes as total sample sizes are fixed by the budget and considering various administrative reasons beforehand, hence incurring extra cost cannot be considered. SAE approach provides a unique way to deal with the small sample sizes problem and even the problem of zero small sizes at micro levels and render stable estimates by utilizing the data from already existing survey (for study variable) and census or administrative records (for auxiliary variables) and quick to produce local level figures without extra budget and time constraints. Basically, it incorporates the idea of “borrowing strength” from related small areas or domains and thus increases the “effective sample sizes” of each domain. SAE techniques are generally model based methods, whereas statistical models are utilized to link the variable of interest with the auxiliary information. The concept of increasing effective sample size combining survey with census information through certain statistical models instead of increasing the sample size, makes the SAE quite distinctive and applicable.

Small area models are special cases of linear mixed models and incorporate random area-specific effects which account for unstructured heterogeneity across areas beyond that explained by auxiliary variables included in the fixed effect part of the model (Rao and Molina, 2015). Based on the level of auxiliary information available, we differentiate between SAE methods based on unit-level models and those based on area-level models. In the former case these models are for the individual survey measurements and include area effects, while in the latter case these models are used to smooth out the variability in the unstable area-level direct survey estimates. Area-level modelling is typically used when unit-level data are unavailable, or, as is often the case, where model covariates are only available at area-level. The Fay–Herriot (FH) model (Fay and Herriot 1979), is a widely used area-level model that incorporates area-specific survey estimates and auxiliary variables, and follow an area-level linear mixed model with independent area random effects. The application of FH model and its various extensions are widely available in various real-life studies and literature to solve the small domain estimation problems (Chandra 2013; Portar et al. 2014, Rao and Molina 2015, Chandra et al. 2018; Anjoy et al. 2018).

In FH model an implicit independence assumption is also imposed on the random effect

component which implies different small areas are simply uncorrelated. However, the spatial dependence between neighbouring areas cannot be denied, since neighbouring districts often influence each other in respect of living standards and life styles. Thus, to incorporate the neighboring effect, it is reasonable to construct a spatial model to capture the spatial association between areas. Therefore, FH model can be extended to incorporate spatially correlated random effects using Conditional Autoregressive Process (CAR) and Simultaneous Autoregressive Process (SAR) specifications. These models define the dependence between areas by using certain contiguity matrix, which can be obtained by using coordinates of the centroid of each small area, its geometric properties (extension, perimeter, etc.) reflecting the neighbourhood structure of adjacent small areas or districts. For the estimation of MPCE, present study considers SAR specifications to incorporate the spatial structure for generating model-based district level estimates.

## 4.2 Small Area Models and estimates

Let there be  $D$  domains (or small areas) labeled  $1, \dots, i, \dots, D$  and let  $\theta_i$  be the parameter to be estimated for domain  $i$ ,  $i = 1, \dots, D$ . Let  $s$  be the chosen sample. Let  $y_i$  denote the unbiased design-based direct survey estimate of the population parameter  $\theta_i$  as obtained from the sample. The simple area specific standard Fay and Herriot (1979) model consisting of a Sampling model and a Linking model is:

$$\textbf{Sampling model: } y_i = \theta_i + e_i$$

where  $e_i$  denotes the independent sampling errors associated with the direct estimator  $y_i$ ;  $e_i$ 's assumed to be independent and identically distributed normal variables with mean 0 and variance  $\sigma_{ei}^2$ ,  $i = 1, \dots, D$ ,  $\sigma_{ei}^2$  being the sampling variance,

and

$$\textbf{Linking model: } \theta_i = \mathbf{x}_i' \boldsymbol{\beta} + v_i$$

where  $\mathbf{x}_i = (1, x_{i1}, \dots, x_{ip-1})'$  denotes the vector of  $p$  known auxiliary variables for area  $i$ , (as obtained from administrative and/or census records)  $\boldsymbol{\beta} = (\beta_0, \dots, \beta_{p-1})'$  is the vector of  $p$  of unknown fixed-effects regression coefficients, and  $v_i$  denotes the area specific random effect independent of  $e_i$ ;  $v_i$  assumed to be independent and identically (i.i.d) distributed normal variables with mean 0 and

variance  $\sigma_v^2$ ,  $\sigma_v^2$  being the model variance. Usually, the sampling variances  $\sigma_{ei}^2, i = 1, \dots, D$  are assumed to be known and these are obtained from survey data considering the underlying survey design.

The FH model implicitly assumes that direct survey estimates from different small areas are uncorrelated. However, in practice there appears to be no good reason why neighbouring areas should not be correlated. In small area modeling, incorporating the information of spatial dependence between neighboring areas, often improves the model accuracy. Therefore, to incorporate spatial dependence in error structure, SAR error process is often used (Molina, I. and Marhuenda, Y. (2015), Anjoy and Chandra, (2023)).

We first give the standard Fay-Herriot (FH) model and then extend it to a spatial model.

Let there be  $D$  domains labeled  $1, \dots, D$  and let  $\theta_d$  be the parameter to be estimated for domain  $d$ ,  $d = 1, \dots, D$ . Let  $s$  be the chosen sample.

#### 6.1. Standard Fay-Herriot Model

(a) Sampling Model:  $\hat{\theta}_d = \theta_d + e_d, d = 1, \dots, D$

where  $\hat{\theta}_d$  is the unbiased direct estimator of  $\theta_d$  from  $s$ ,  $e_d$  denotes independent sampling errors,  $e_d \sim iid N(0, \sigma_d^2)$ ,  $\sigma_d^2$  is sampling variance.

(b) Linking Model:  $\theta_d = x_d' \beta + u_d, d = 1, \dots, D$

where  $\beta$  is the vector of unknown fixed-effect parameters,  $u_d$  is a random effect independent of  $e_d$ ,  $u_d \sim iid N(0, \sigma_u^2)$ ,  $\sigma_u^2$  is model variance,  $x_d$  is the  $p \times 1$  known vector of domain-specific covariates.

Combining models (a) and (b) we get the Fay-Herriot (FH) model as:

$$\hat{\theta} = \mathbf{X}\beta + \mathbf{u} + \mathbf{e} \quad (1)$$

where  $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_D)'$ ,  $\mathbf{e}$  and  $\mathbf{u}$  defined similarly,  $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_D)'$ . So,

$\hat{\theta} \sim N(\mathbf{X}\beta, \Sigma(\sigma_u^2))$ , where  $\Sigma(\sigma_u^2) = \text{Diag}(\sigma_1^2 + \sigma_u^2, \dots, \sigma_D^2 + \sigma_u^2)$ .

Let  $\hat{\beta}$  be the weighted least squares estimator for  $\beta$  and  $\hat{u}_d$  the predicted random effects. Then the Best Linear Unbiased Predictor (BLUP) of  $\theta_d$  is

$$\hat{\theta}_d^{EBLUP}(FH) = \hat{x}_d' \hat{\beta} + \hat{u}_d = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_d^2} \hat{\theta}_d + \frac{\sigma_d^2}{\sigma_u^2 + \sigma_d^2} x_d' \hat{\beta}.$$

Then the empirical BLUP (EBLUP) of  $\theta_d$  is



$$\hat{\theta}_d^{EBLUP}(FH) = \frac{\sigma_u^2}{\sigma_u^2 + \sigma_d^2} \hat{\theta}_d + \frac{\sigma_d^2}{\sigma_u^2 + \sigma_d^2} x_d' \hat{\beta},$$

where we used REML estimates  $\hat{\beta}$  and  $\sigma_u^2$  and bootstrap estimates  $\hat{\sigma}_d^2$ . For a  $d$  with no sample units,  $\hat{\theta}_{FH}^{EBLUP} = x_d' \hat{\beta}$ .

## 6.2. Spatial Fay-Herriot (SFH) Model

The FH model (1) assumes the random effects to be independent. Now we assume correlated random effects to incorporate a certain neighboring structure. For this, we consider that  $u$  in the equation (1) follows a simultaneously autoregressive process of order one, SAR(1), with unknown spatial autoregressive coefficient  $\rho \in (-1, 1)$ . The  $u$  is replaced by

$$\mathbf{u} = \rho \mathbf{W} \mathbf{u} + \mathbf{v}, \quad (2)$$

where  $u$  is the vector of area effects,  $W$  is the  $D \times D$  row-standardized proximity matrix describing the neighborhood structure,  $\mathbf{v} \sim N(0_D, \sigma_v^2 \mathbf{I}_D)$  for unknown  $\sigma_v^2$ . Then from (1) and (2), the spatial FH model can be written as

$$\hat{\theta} = \mathbf{X}\beta + (\mathbf{I}_D - \rho W)^{-1} \mathbf{v} + \mathbf{e}.$$

As  $\mathbf{e}$  is independent of  $\mathbf{v}$ , the covariance matrix of  $\hat{\theta}$  is equal to  $\Sigma = \sigma_v^2 [(\mathbf{I}_D - \rho W)'(\mathbf{I}_D - \rho W)]^{-1} + \text{diag}(\sigma_1^2, \dots, \sigma_D^2)$ , a function of  $\sigma_v^2, \rho$ .

The spatial BLUP of  $\theta_d$  is given by

$$\hat{\theta}_d^{EBLUP}(SFH) = x_d' \hat{\beta} + b_d' \sigma_v^2 [(\mathbf{I}_D - \rho W)'(\mathbf{I}_D - \rho W)]^{-1} \Sigma^{-1} (\hat{\theta}_d - \mathbf{X} \hat{\beta}),$$

where  $\hat{\beta}$  is the weighted least squares estimator of  $\beta$  and  $b_d$  is a  $D \times 1$  vector with 1 in the  $d$ -th position and 0 elsewhere.

Here,  $\hat{\beta}$  and  $\Sigma$  are functions of the unknown  $\sigma_v^2$  and  $\rho$ . Replacing the unknown parameters in the BLUP by their consistent estimators, we get the EBLUP of  $\theta_d$  as

$$\hat{\theta}_d^{EBLUP}(SFH) = x_d' \hat{\beta} + b_d' \hat{\sigma}_v^2 [(\mathbf{I}_D - \hat{\rho} W)'(\mathbf{I}_D - \hat{\rho} W)]^{-1} \hat{\Sigma}^{-1} (\hat{\theta}_d - \mathbf{X} \hat{\beta}).$$

# **CHAPTER 5**

## **FINDINGS AND ANALYSIS**

## 5 Results and Discussion

In this study, direct estimates  $y_i$  is the MPCE estimates for district  $i$  based on the survey design. For Rural and Urban sectors analysis were done separately based on data from HCES 2022-23. From this survey, for the State of Uttar Pradesh information on direct MPCE were available for 71 districts which is same as no. of districts classified according to previous Census. Whereas, currently, the state has 75 districts. Therefore, attempt was made to obtain auxiliary variables for all the districts, to ensure small area model-based MPCE estimates for present 75 districts. In case of newly formed and restructured districts, some approximations were used to get the direct survey-based district-wise MPCE for Rural and Urban sectors separately. Such reallocation has affected mainly the new districts Amethi, Hapur, Shambhal, Shamli districts. If HCES data were available with FSUs classified into 75 districts, then these approximations would not have been necessary.

Urban sector of Shravasti district did not receive any FSU and so direct estimate could not be obtained. An estimate for Shravasti Urban was obtained via model-based prediction.

The variance estimates for each district were bootstrap-based variance estimates. The available HCES 2022-23 data does not give the detailed weights, instead a consolidated multiplier is provided. So, the variance of the direct estimates for the districts could not be obtained using the exact formula-wise and stage-wise sampling weights. Therefore, bootstrap estimates of variance were used. If detailed weights had been available, the design-based variance estimates could be obtained. It may be mentioned that bootstrap-based variance estimates are a common method used to assess the variability of estimates, particularly in situations where traditional variance estimation methods may be challenging, such as when dealing with complex survey designs or limited data within specific subgroups or strata.

For contiguity matrix, several approaches were tried. e.g., districts with common boundaries, distance between districts, districts within same Mandals. Finally, districts sharing common boundaries were considered as 'neighbours'.

Availability of good auxiliary data is crucial to obtain improved estimates. Extensive exploration on pool of auxiliary variables were done. Finally, based on substantial correlation with the direct survey estimates, the following covariates were chosen for modeling:

**Rural:** Number of Beneficiaries for Old age pension (Samaj Kalyan Vibhag) and Number of patients covered in Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB PM-JAY)

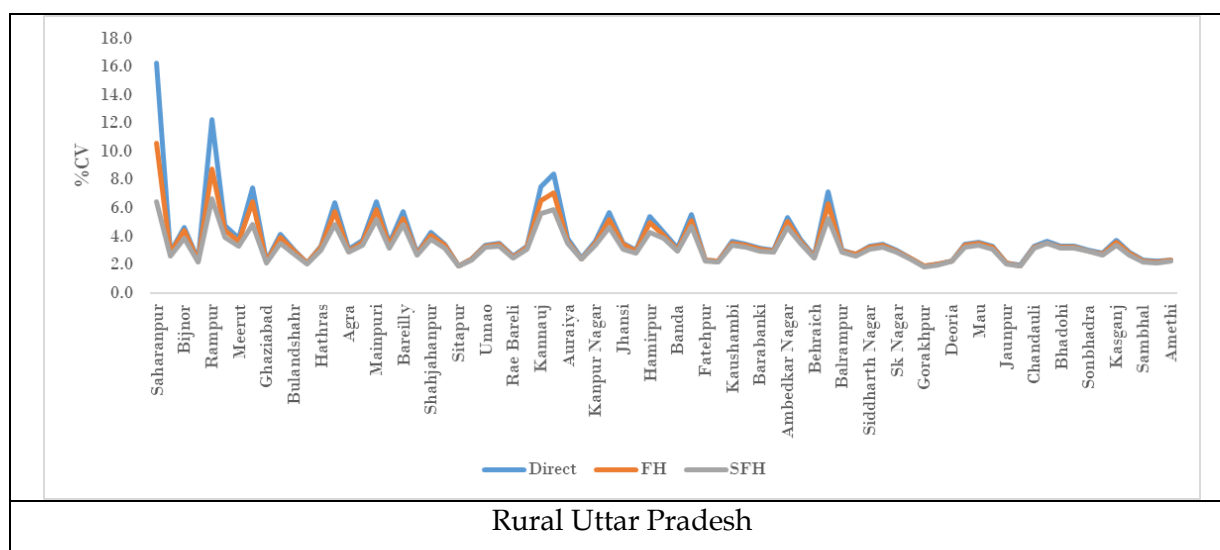
**Urban:** Number of households using Domestic LPG connection, Number of Beneficiaries and food distributed (in Ton) to Antyodaya card holders.

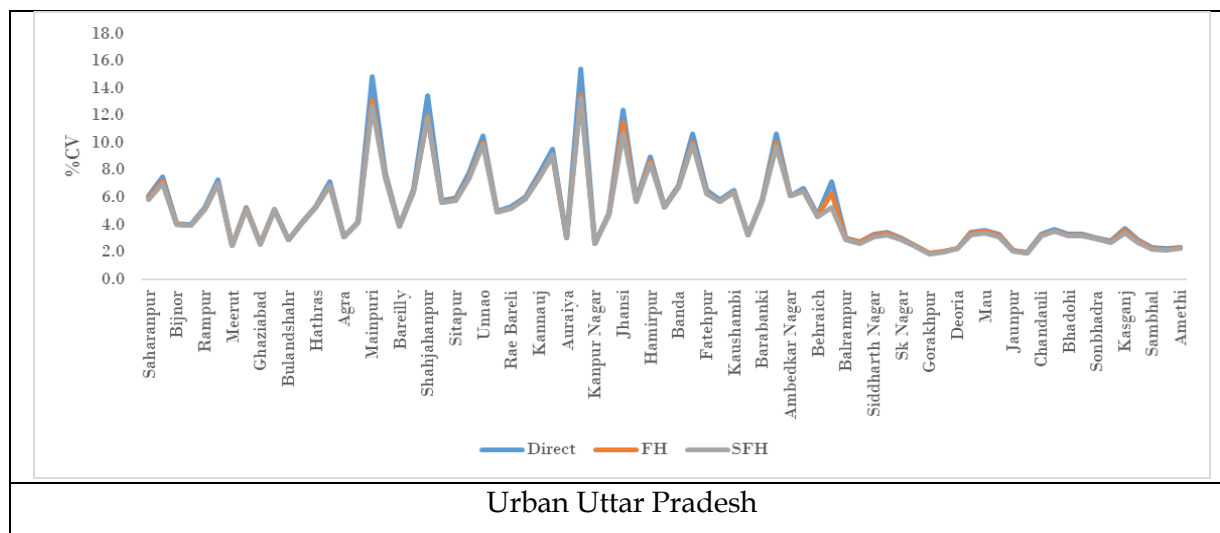
In addition, spatial modeling was used to see if `neighbours' affected the MPCE values.

The small area model-based estimates of MPCE were produced using both the FH and SFH model. Traditional direct survey estimates for MPCE were also computed to carry out comparison of small area model-based estimates vs. direct estimation approach. Percentage coefficient of variation (% CV) is the criteria which have been used to indicate the better performing model with stable estimates. Table 2 and Table 3 in the Appendix present Direct and model-based small area estimates produced by FH, SFH approach along with their %CV for Rural and Urban sectors respectively. It may be seen from Fig. 1 also that the %CV for estimates based on SAE were less than the corresponding %CV for direct estimates. Thus, SAE approach gave improved estimates.

Estimates for the urban part of the district with zero sample size could also be obtained.

The %CV were further reduced for estimates obtained through spatial modeling for most of the districts using SFH. This reduction was generally more in the rural sectors than in the urban sectors.





**Fig 1: District-wise %Coefficient of Variation Generated based on Direct and Model based SAE Approach**

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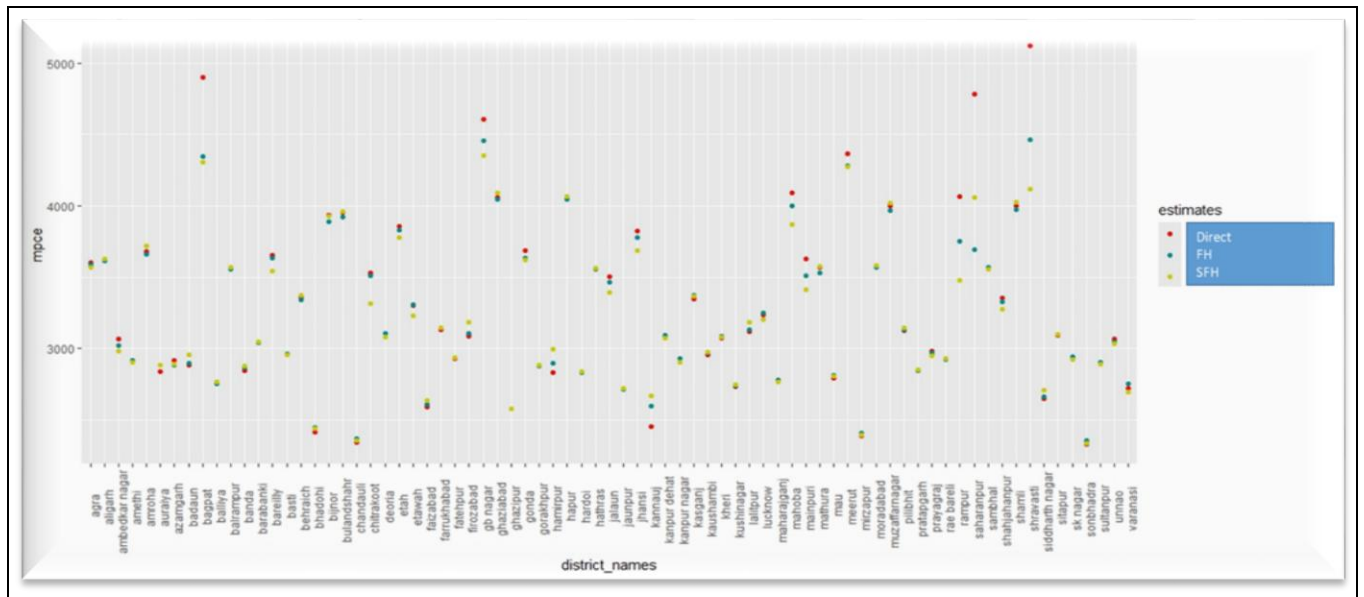


Fig. 2: District-wise Direct vs. Model-based MPCE Estimates for Rural Sector

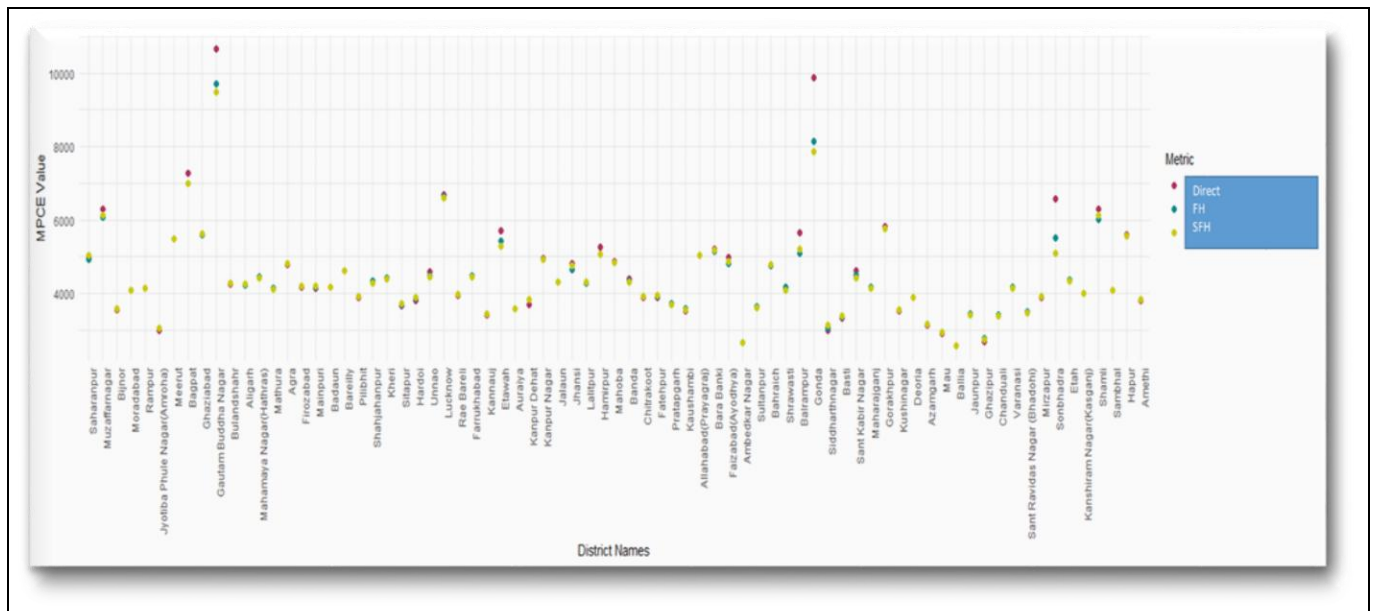


Fig. 3: District-wise Direct vs. Model-based MPCE Estimates for Urban Sector

## Findings

1. Estimates could be obtained separately for rural and urban parts of each district, based on small-area estimation (SAE) methodology.
2. The RSE for estimates based on SAE were less than the corresponding RSE for direct estimates for rural parts of all districts and urban parts of all districts except two. Thus, modeling gave improved estimates.
3. Estimates for the urban part of the district with zero sample size could also be obtained. This was done via model prediction.

4. The RSE's were further reduced for estimates obtained through spatial modeling for most districts. This reduction was generally more in the rural sectors than in the urban sectors.

### **Some Remarks**

1. Availability of good auxiliary data is crucial to obtaining good estimates. From the data given by UP DES, 3 'good' covariates were chosen for the urban and 2 for rural parts, for illustrating the proposed methodology. With some other covariates, these estimates might be improved.

2. For 'borrowing strength' from neighbouring areas, we tried several approaches, e.g., districts with common boundaries, distance between districts, districts within same Mandals. Finally, districts within common boundaries as 'neighbours' were found to be better and so this was used in our computations.

3. Further modifications might be possible. However, due to time constraints, this was not explored.

4. This approach can be applied to other surveys as well.

### **Two caveats and possible modifications**

The data, as available for HCES 22-23, necessitated two approximations in our computations as listed below:

1. The available HCES 22-23 data classifies the fsu's into 71 districts in UP. So, if we had wanted direct MPCE estimates for only these 71 districts, no approximations would have been needed. However, with 75 districts now, direct MPCE estimates for the changed districts could not be obtained from the information available. So, we used some approximations to get these estimates for the changed districts. If HCES data files are available with UP fsu's classified into 75 districts, then revised estimates can be obtained.

2. The available HCES 22-23 data does not give the detailed break-up of the weights, instead a consolidated multiplier is given. So, the variance of the direct estimates for the districts could not be obtained using the exact formula-wise and stage-wise sampling weights. Instead, bootstrap estimates of variance were used. If detailed weights are available, the design-based variance estimates can be obtained to get the RSEs.

# **CHAPTER 6**

## **Conclusions and Way forward**



## 6 Conclusions and policy implications

The purpose of survey sampling is to obtain statistical information about a finite population by selecting a probability sample from this population, measuring the required information about the units in this sample and estimating finite population parameters such as mean, total, ratio etc. Although, increasing demand for high-resolution, timely, and policy-relevant data—particularly at local levels such as districts, blocks, or villages—has made model-based estimation a valuable complement to traditional survey methods. While design-based inference remains the cornerstone of official statistics due to its transparency and robustness, it often faces limitations in domains with small sample sizes or high variability, where direct estimates become unreliable or unusable. In the design-based and model-assisted approach, the statistical inference is based on the stochastic structure induced by the sampling design. Parameter and variance estimators are derived under the concept of repeatedly drawing samples from a finite population according to the same sampling design, while statistical modelling plays a minor role. By leveraging auxiliary data and statistical models, the model-based approach allows for:

- **Stable and reliable estimates in small domains**, even when direct survey data is sparse.
- **Continuity in statistical series** during survey redesigns, changes in data collection modes, or breaks in time series—addressing a key concern for long-term policy evaluation.
- The **expansion of estimation capabilities** to previously unreachable levels of disaggregation, enabling more inclusive and localized policy planning.

In the model-based context, the probability structure of the sampling design plays a less pronounced role, since the inference is based on the probability structure of an assumed statistical model. Model-based procedures can be adopted in situations where direct estimators do not result in sufficiently reliable estimates. However, model-based estimates should be clearly separated from the design-based ones, and accompanied with appropriate methodology and quality descriptions, where the underlying model assumptions are made explicit.

Survey redesigns and survey transitions generally affect the continuity of series. Model-based approaches can be useful to quantify the effect of a redesign and to meet the users' requirement of having uninterrupted series. Thus, the MPCE generated at district level utilizing the suggested method of model-based estimates can be effectively used to meet this requirement of the users and can fill critical gaps by providing granular insights into living standards and consumption patterns across districts and sub-districts. These estimates can inform the design and targeting of welfare schemes, poverty alleviation programs, and regional development initiatives. Moreover, the model-based framework is scalable and flexible and utilizing it for future applications after identification of appropriate covariates for estimating a wide range of socio-economic indicators:

- Estimation of MPCE at further disaggregated level, namely, at block or village level
- Estimation of other similar kind of parameters, like, LFPR, WPR, UR etc.

Model-based estimation for smaller domains can be a powerful tool for taking data-driven and informed policy decisions across various domains. However, the limitations and potential biases of models and consequences arising out of these are required to be acknowledged and considered to ensure that model-based estimation serves the intended purpose in policymaking:

- Model accuracy is highly contingent on the quality and relevance of covariates used.
- Estimates are sensitive to model assumptions, and misspecification can lead to misleading results.
- Users must be clearly informed about the nature of these estimates, including uncertainty measures, and distinctions from design-based statistics.

Thus, while model-based estimation is not a replacement for direct estimation, it serves as a critical supplement, especially in addressing the growing demand for data disaggregation and continuity in an evolving statistical landscape.

## 7 Way Forward

Model-based estimation involves a strategic focus on methodological rigor, institutional integration, and effective communication. A key step is the identification and integration of appropriate covariates to enhance the explanatory power of models. Auxiliary information from sources such as administrative records, census data, and remote sensing should be leveraged. Covariates like household characteristics, asset ownership, education levels, and employment

indicators can significantly improve the precision of estimates such as Monthly Per Capita Expenditure (MPCE) and other related parameters.

Equally important is the development of standardized model frameworks that can be replicated across different geographic levels. These models should be subject to rigorous validation procedures, including cross-validation and benchmarking against available direct survey estimates, to assess their reliability and performance.

Institutionalizing model-based estimation within the official statistical system is essential for its long-term sustainability. This includes positioning it as a complementary approach to traditional design-based estimation, particularly in the context of small area estimation (SAE). To ensure transparency and credibility, all model documentation should clearly outline assumptions, data sources, and quality measures such as coefficients of variation (CV), coverage rates etc..

Transparent communication with data users—especially policymakers and planners—is crucial. Users must be informed about the model-based nature of estimates, the associated uncertainty (e.g., credible intervals), and any potential biases or limitations inherent in the modeling approach. Such openness fosters trust in the estimates and facilitates evidence-based decision-making.

To support implementation, capacity building is vital. Statistical personnel and analysts should be trained in small area estimation techniques and model diagnostics. Collaboration among statistical agencies, academic institutions, and data scientists can further enhance both the methodology and its applications. Starting with pilot implementations in selected districts or states allows for refinement of the models before scaling up to national coverage. Throughout this process, models should remain adaptable to regional data availability and specific policy needs.

## Appendix A

| List of Tables |                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Table No.      | Title                                                                                                                             |
| A1             | <b>District wise Number of FSUs surveyed in HCES:2022-23 (71 districts)</b>                                                       |
| A2             | <b>District wise revised number of allocated FSUs among 75 districts</b>                                                          |
| A3             | <b>District-wise MPCE Estimates and %Coefficient of Variation (CV) using Direct and Model-based SAE Approach for Rural Sector</b> |
| A4             | <b>District-wise MPCE Estimates and %Coefficient of Variation (CV) using Direct and Model-based SAE Approach for Urban Sector</b> |

**Table A1: District wise Number of FSUs surveyed**

| (1)<br>SNo. | (2)<br>District        | (3)<br>Rural | (4)<br>Urban | (5)<br>Rural + Urban |
|-------------|------------------------|--------------|--------------|----------------------|
| 1           | Agra                   | 19           | 23           | 42                   |
| 2           | Aligarh                | 22           | 15           | 37                   |
| 3           | Allahabad              | 38           | 29           | 67                   |
| 4           | Ambedkar Nagar         | 12           | 2            | 14                   |
| 5           | Amethi*                | 0            | 0            | 0                    |
| 6           | Auraiya                | 9            | 3            | 12                   |
| 7           | Azamgarh               | 30           | 6            | 36                   |
| 8           | Baghpat                | 6            | 4            | 10                   |
| 9           | Bahraich               | 23           | 6            | 29                   |
| 10          | Ballia                 | 21           | 3            | 24                   |
| 11          | Balrampur              | 13           | 1            | 14                   |
| 12          | Banda                  | 15           | 3            | 18                   |
| 13          | Bara Banki             | 24           | 6            | 30                   |
| 14          | Bareilly               | 20           | 26           | 46                   |
| 15          | Basti                  | 26           | 1            | 27                   |
| 16          | Bijnor                 | 15           | 9            | 24                   |
| 17          | Budaun                 | 25           | 3            | 28                   |
| 18          | Bulandshahr            | 12           | 8            | 20                   |
| 19          | Chandauli              | 16           | 3            | 19                   |
| 20          | Chitrakoot             | 5            | 1            | 6                    |
| 21          | Deoria                 | 20           | 3            | 23                   |
| 22          | Etah                   | 8            | 5            | 13                   |
| 23          | Etawah                 | 4            | 4            | 8                    |
| 24          | Faizabad               | 14           | 2            | 16                   |
| 25          | Farrukhabad            | 10           | 7            | 17                   |
| 26          | Fatehpur               | 22           | 3            | 25                   |
| 27          | Firozabad              | 8            | 4            | 12                   |
| 28          | Gautam Buddha<br>Nagar | 10           | 39           | 49                   |
| 29          | Ghaziabad              | 17           | 43           | 60                   |
| 30          | Ghazipur               | 26           | 1            | 27                   |
| 31          | Gonda                  | 29           | 4            | 33                   |
| 32          | Gorakhpur              | 18           | 12           | 30                   |
| 33          | Hamirpur               | 2            | 3            | 5                    |
| 34          | Hapur*                 | 0            | 0            | 0                    |
| 35          | Hardoi                 | 28           | 3            | 31                   |
| 36          | Jalaun                 | 7            | 4            | 11                   |
| 37          | Jaunpur                | 24           | 9            | 33                   |
| 38          | Jhansi                 | 8            | 14           | 22                   |
| 39          | Jyotiba Phule Nagar    | 9            | 2            | 11                   |
| 40          | Kannauj                | 6            | 4            | 10                   |
| 41          | Kanpur Dehat           | 17           | 1            | 18                   |
| 42          | Kanpur Nagar           | 12           | 45           | 57                   |
| 43          | Kanshiram Nagar        | 3            | 3            | 6                    |
| 44          | Kaushambi              | 14           | 2            | 16                   |

| <b>Table A1: District wise Number of FSUs surveyed</b> |                                 |                            |                            |                                    |
|--------------------------------------------------------|---------------------------------|----------------------------|----------------------------|------------------------------------|
| <b>(1)</b><br><b>SNo.</b>                              | <b>(2)</b><br><b>District</b>   | <b>(3)</b><br><b>Rural</b> | <b>(4)</b><br><b>Urban</b> | <b>(5)</b><br><b>Rural + Urban</b> |
| 45                                                     | Kheri                           | 19                         | 5                          | 24                                 |
| 46                                                     | Kushinagar                      | 25                         | 4                          | 29                                 |
| 47                                                     | Lalitpur                        | 8                          | 4                          | 12                                 |
| 48                                                     | Lucknow                         | 14                         | 43                         | 57                                 |
| 49                                                     | Mahamaya Nagar                  | 8                          | 7                          | 15                                 |
| 50                                                     | Mahoba                          | 2                          | 2                          | 4                                  |
| 51                                                     | Mahrajanj                       | 19                         | 1                          | 20                                 |
| 52                                                     | Mainpuri                        | 7                          | 3                          | 10                                 |
| 53                                                     | Mathura                         | 10                         | 4                          | 14                                 |
| 54                                                     | Mau                             | 9                          | 7                          | 16                                 |
| 55                                                     | Meerut                          | 11                         | 26                         | 37                                 |
| 56                                                     | Mirzapur                        | 12                         | 6                          | 18                                 |
| 57                                                     | Moradabad                       | 25                         | 12                         | 37                                 |
| 58                                                     | Muzaffarnagar                   | 22                         | 11                         | 33                                 |
| 59                                                     | Pilibhit                        | 10                         | 5                          | 15                                 |
| 60                                                     | Pratapgarh                      | 27                         | 4                          | 31                                 |
| 61                                                     | Rae Bareli                      | 24                         | 3                          | 27                                 |
| 62                                                     | Rampur                          | 8                          | 15                         | 23                                 |
| 63                                                     | Saharanpur                      | 11                         | 14                         | 25                                 |
| 64                                                     | Sambhal*                        | 0                          | 0                          | 0                                  |
| 65                                                     | Sant Kabir Nagar                | 13                         | 2                          | 15                                 |
| 66                                                     | Sant Ravidas Nagar<br>(Bhadohi) | 13                         | 3                          | 16                                 |
| 67                                                     | Shahjahanpur                    | 20                         | 2                          | 22                                 |
| 68                                                     | Shamli*                         | 0                          | 0                          | 0                                  |
| 69                                                     | Shrawasti                       | 3                          | 0                          | 3                                  |
| 70                                                     | Siddharthnagar                  | 11                         | 3                          | 14                                 |
| 71                                                     | Sitapur                         | 38                         | 3                          | 41                                 |
| 72                                                     | Sonbhadra                       | 13                         | 4                          | 17                                 |
| 73                                                     | Sultanpur                       | 19                         | 6                          | 25                                 |
| 74                                                     | Unnao                           | 24                         | 3                          | 27                                 |
| 75                                                     | Varanasi                        | 16                         | 29                         | 45                                 |
|                                                        | <b>Grand Total</b>              | <b>1108</b>                | <b>600</b>                 | <b>1708</b>                        |

**Table A2: District wise revised number of allocated FSUs among 75 districts**

| (1)<br>SNo. | (2)<br>District        | (3)<br>Rural | (4)<br>Urban | (5)<br>Rural + Urban |
|-------------|------------------------|--------------|--------------|----------------------|
| 1           | Agra                   | 19           | 23           | 42                   |
| 2           | Aligarh                | 22           | 15           | 37                   |
| 3           | Allahabad              | 38           | 29           | 67                   |
| 4           | Ambedkar Nagar         | 12           | 2            | 14                   |
| 5           | Amethi*                | 14           | 4            | 18                   |
| 6           | Auraiya                | 9            | 3            | 12                   |
| 7           | Azamgarh               | 30           | 6            | 36                   |
| 8           | Baghpat                | 6            | 4            | 10                   |
| 9           | Bahraich               | 23           | 6            | 29                   |
| 10          | Ballia                 | 21           | 3            | 24                   |
| 11          | Balrampur              | 13           | 1            | 14                   |
| 12          | Banda                  | 15           | 3            | 18                   |
| 13          | Bara Banki             | 24           | 6            | 30                   |
| 14          | Bareilly               | 20           | 26           | 46                   |
| 15          | Basti                  | 26           | 1            | 27                   |
| 16          | Bijnor                 | 15           | 9            | 24                   |
| 17          | Budaun                 | 21           | 3            | 24                   |
| 18          | Bulandshahr            | 12           | 8            | 20                   |
| 19          | Chandauli              | 16           | 3            | 19                   |
| 20          | Chitrakoot             | 5            | 1            | 6                    |
| 21          | Deoria                 | 20           | 3            | 23                   |
| 22          | Etah                   | 8            | 5            | 13                   |
| 23          | Etawah                 | 4            | 4            | 8                    |
| 24          | Faizabad               | 14           | 2            | 16                   |
| 25          | Farrukhabad            | 10           | 7            | 17                   |
| 26          | Fatehpur               | 22           | 3            | 25                   |
| 27          | Firozabad              | 8            | 4            | 12                   |
| 28          | Gautam Buddha<br>Nagar | 10           | 39           | 49                   |
| 29          | Ghaziabad              | 1            | 36           | 37                   |
| 30          | Ghazipur               | 26           | 1            | 27                   |
| 31          | Gonda                  | 29           | 4            | 33                   |
| 32          | Gorakhpur              | 18           | 12           | 30                   |
| 33          | Hamirpur               | 2            | 3            | 5                    |
| 34          | Hapur*                 | 16           | 7            | 23                   |
| 35          | Hardoi                 | 28           | 3            | 31                   |
| 36          | Jalaun                 | 7            | 4            | 11                   |
| 37          | Jaunpur                | 24           | 9            | 33                   |
| 38          | Jhansi                 | 8            | 14           | 22                   |
| 39          | Jyotiba Phule Nagar    | 9            | 2            | 11                   |
| 40          | Kannauj                | 6            | 4            | 10                   |
| 41          | Kanpur Dehat           | 17           | 1            | 18                   |
| 42          | Kanpur Nagar           | 12           | 45           | 57                   |
| 43          | Kanshiram Nagar        | 3            | 3            | 6                    |
| 44          | Kaushambi              | 14           | 2            | 16                   |
| 45          | Kheri                  | 19           | 5            | 24                   |
| 46          | Kushinagar             | 25           | 4            | 29                   |
| 47          | Lalitpur               | 8            | 4            | 12                   |
| 48          | Lucknow                | 14           | 43           | 57                   |

| <b>Table A2: District wise revised number of allocated FSUs among 75 districts</b> |                                 |                            |                            |                                    |
|------------------------------------------------------------------------------------|---------------------------------|----------------------------|----------------------------|------------------------------------|
| <b>(1)</b><br><b>SNo.</b>                                                          | <b>(2)</b><br><b>District</b>   | <b>(3)</b><br><b>Rural</b> | <b>(4)</b><br><b>Urban</b> | <b>(5)</b><br><b>Rural + Urban</b> |
| 49                                                                                 | Mahamaya Nagar                  | 8                          | 7                          | 15                                 |
| 50                                                                                 | Mahoba                          | 2                          | 2                          | 4                                  |
| 51                                                                                 | Mahrajganj                      | 19                         | 1                          | 20                                 |
| 52                                                                                 | Mainpuri                        | 7                          | 3                          | 10                                 |
| 53                                                                                 | Mathura                         | 10                         | 4                          | 14                                 |
| 54                                                                                 | Mau                             | 9                          | 7                          | 16                                 |
| 55                                                                                 | Meerut                          | 11                         | 26                         | 37                                 |
| 56                                                                                 | Mirzapur                        | 12                         | 6                          | 18                                 |
| 57                                                                                 | Moradabad                       | 12                         | 9                          | 21                                 |
| 58                                                                                 | Muzaffarnagar                   | 15                         | 8                          | 23                                 |
| 59                                                                                 | Pilibhit                        | 10                         | 5                          | 15                                 |
| 60                                                                                 | Pratapgarh                      | 27                         | 4                          | 31                                 |
| 61                                                                                 | Rae Bareli                      | 19                         | 3                          | 22                                 |
| 62                                                                                 | Rampur                          | 8                          | 15                         | 23                                 |
| 63                                                                                 | Saharanpur                      | 11                         | 14                         | 25                                 |
| 64                                                                                 | Sambhal*                        | 17                         | 3                          | 20                                 |
| 65                                                                                 | Sant Kabir Nagar                | 13                         | 2                          | 15                                 |
| 66                                                                                 | Sant Ravidas Nagar<br>(Bhadohi) | 13                         | 3                          | 16                                 |
| 67                                                                                 | Shahjahanpur                    | 20                         | 2                          | 22                                 |
| 68                                                                                 | Shamli*                         | 7                          | 3                          | 10                                 |
| 69                                                                                 | Shrawasti                       | 3                          |                            | 3                                  |
| 70                                                                                 | Siddharthnagar                  | 11                         | 3                          | 14                                 |
| 71                                                                                 | Sitapur                         | 38                         | 3                          | 41                                 |
| 72                                                                                 | Sonbhadra                       | 13                         | 4                          | 17                                 |
| 73                                                                                 | Sultanpur                       | 10                         | 2                          | 12                                 |
| 74                                                                                 | Unnao                           | 24                         | 3                          | 27                                 |
| 75                                                                                 | Varanasi                        | 16                         | 29                         | 45                                 |
|                                                                                    | <b>Grand Total</b>              | <b>1108</b>                | <b>600</b>                 | <b>1708</b>                        |

**Note:** \* Newly created districts



**Table A3: District-wise MPCE Estimates and %Coefficient of Variation (CV) using Direct and Model-based SAE Approach for Rural Sector**

| (1)           | (2)            |      |      | (3)    |      |     |
|---------------|----------------|------|------|--------|------|-----|
| District      | MPCE Estimates |      |      | %CV    |      |     |
|               | Direct         | FH   | SFH  | Direct | FH   | SFH |
| Saharanpur    | 4781           | 3692 | 4056 | 16.3   | 10.6 | 6.5 |
| Muzaffarnagar | 3999           | 3967 | 4020 | 2.9    | 2.8  | 2.6 |
| Bijnor        | 3932           | 3892 | 3930 | 4.7    | 4.4  | 3.9 |
| Moradabad     | 3566           | 3572 | 3582 | 2.3    | 2.3  | 2.2 |
| Rampur        | 4062           | 3750 | 3478 | 12.3   | 8.7  | 6.7 |
| Amroha        | 3683           | 3661 | 3722 | 4.8    | 4.5  | 3.9 |
| Meerut        | 4362           | 4279 | 4272 | 3.9    | 3.7  | 3.3 |
| Bagpat        | 4899           | 4347 | 4306 | 7.4    | 6.5  | 4.9 |
| Ghaziabad     | 4058           | 4044 | 4092 | 2.2    | 2.2  | 2.1 |
| Gb Nagar      | 4606           | 4454 | 4353 | 4.1    | 3.9  | 3.5 |
| Bulandshahr   | 3953           | 3922 | 3962 | 3      | 2.9  | 2.8 |
| Aligarh       | 3621           | 3614 | 3626 | 2.1    | 2.1  | 2   |
| Hathras       | 3562           | 3559 | 3561 | 3.3    | 3.2  | 3   |
| Mathura       | 3567           | 3531 | 3578 | 6.4    | 5.8  | 4.8 |
| Agra          | 3605           | 3590 | 3566 | 3.1    | 3    | 2.9 |
| Firozabad     | 3086           | 3105 | 3182 | 3.8    | 3.6  | 3.3 |
| Mainpuri      | 3630           | 3509 | 3415 | 6.5    | 5.9  | 5.2 |
| Badaun        | 2883           | 2899 | 2955 | 3.5    | 3.4  | 3.2 |
| Bareilly      | 3653           | 3633 | 3545 | 5.7    | 5.3  | 4.8 |
| Pilibhit      | 3127           | 3135 | 3147 | 2.8    | 2.8  | 2.7 |
| Shahjahanpur  | 3355           | 3327 | 3275 | 4.3    | 4.1  | 3.8 |
| Kheri         | 3077           | 3087 | 3081 | 3.5    | 3.3  | 3.2 |
| Sitapur       | 3091           | 3099 | 3097 | 1.9    | 1.9  | 1.9 |
| Hardoi        | 2832           | 2831 | 2841 | 2.4    | 2.4  | 2.4 |
| Unnao         | 3066           | 3049 | 3032 | 3.4    | 3.3  | 3.2 |
| Lucknow       | 3235           | 3248 | 3206 | 3.5    | 3.4  | 3.3 |
| Rae Bareli    | 2925           | 2926 | 2931 | 2.6    | 2.5  | 2.5 |
| Farrukhabad   | 3132           | 3146 | 3144 | 3.3    | 3.2  | 3.1 |
| Kannauj       | 2454           | 2597 | 2667 | 7.5    | 6.5  | 5.6 |
| Etawah        | 3299           | 3310 | 3228 | 8.4    | 7.1  | 5.9 |
| Auraiya       | 2838           | 2884 | 2883 | 3.9    | 3.7  | 3.5 |
| Kanpur Dehat  | 3091           | 3094 | 3077 | 2.4    | 2.4  | 2.4 |
| Kanpur Nagar  | 2904           | 2929 | 2907 | 3.7    | 3.5  | 3.4 |
| Jalaun        | 3503           | 3464 | 3393 | 5.7    | 5.2  | 4.7 |

**Table A3: District-wise MPCE Estimates and %Coefficient of Variation (CV) using Direct and Model-based SAE Approach for Rural Sector**

| (1)             | (2)            |      |      | (3)    |     |     |
|-----------------|----------------|------|------|--------|-----|-----|
| District        | MPCE Estimates |      |      | %CV    |     |     |
|                 | Direct         | FH   | SFH  | Direct | FH  | SFH |
| Jhansi          | 3820           | 3777 | 3690 | 3.5    | 3.4 | 3.1 |
| Lalitpur        | 3119           | 3133 | 3186 | 3      | 2.9 | 2.8 |
| Hamirpur        | 2832           | 2896 | 2994 | 5.4    | 5   | 4.3 |
| Mahoba          | 4093           | 3999 | 3868 | 4.3    | 4.1 | 3.9 |
| Banda           | 2848           | 2865 | 2876 | 3.1    | 3.1 | 3   |
| Chitrakoot      | 3531           | 3510 | 3314 | 5.5    | 5.1 | 4.7 |
| Fatehpur        | 2930           | 2936 | 2935 | 2.3    | 2.3 | 2.3 |
| Pratapgarh      | 2849           | 2847 | 2853 | 2.3    | 2.3 | 2.2 |
| Kaushambi       | 2955           | 2971 | 2975 | 3.7    | 3.5 | 3.4 |
| Prayagraj       | 2985           | 2967 | 2948 | 3.4    | 3.4 | 3.2 |
| Barabanki       | 3046           | 3041 | 3050 | 3.2    | 3.1 | 3   |
| Faizabad        | 2593           | 2609 | 2639 | 3      | 3   | 2.9 |
| Ambedkar Nagar  | 3064           | 3019 | 2980 | 5.3    | 5.1 | 4.6 |
| Sultanpur       | 2897           | 2904 | 2893 | 3.8    | 3.7 | 3.5 |
| Behraich        | 3359           | 3340 | 3375 | 2.6    | 2.6 | 2.5 |
| Shrawasti       | 5124           | 4462 | 4118 | 7.2    | 6.3 | 5.2 |
| Balrampur       | 3561           | 3559 | 3572 | 3.1    | 3   | 2.9 |
| Gonda           | 3684           | 3636 | 3622 | 2.8    | 2.7 | 2.6 |
| Siddharth Nagar | 2650           | 2666 | 2706 | 3.3    | 3.2 | 3.1 |
| Basti           | 2961           | 2961 | 2959 | 3.4    | 3.4 | 3.2 |
| Sk Nagar        | 2929           | 2943 | 2922 | 3      | 3   | 2.9 |
| Maharajganj     | 2776           | 2778 | 2770 | 2.5    | 2.5 | 2.4 |
| Gorakhpur       | 2878           | 2878 | 2883 | 1.9    | 1.9 | 1.9 |
| Kushinagar      | 2736           | 2739 | 2746 | 2.1    | 2   | 2   |
| Deoria          | 3104           | 3105 | 3079 | 2.3    | 2.3 | 2.2 |
| Azamgarh        | 2917           | 2885 | 2891 | 3.5    | 3.4 | 3.3 |
| Mau             | 2794           | 2814 | 2808 | 3.6    | 3.5 | 3.4 |
| Balliya         | 2753           | 2753 | 2765 | 3.3    | 3.2 | 3.1 |
| Jaunpur         | 2719           | 2713 | 2718 | 2.1    | 2.1 | 2.1 |
| Ghazipur        | 2577           | 2579 | 2579 | 2      | 1.9 | 1.9 |
| Chandauli       | 2343           | 2367 | 2353 | 3.3    | 3.3 | 3.2 |
| Varanasi        | 2723           | 2755 | 2697 | 3.7    | 3.5 | 3.5 |
| Bhadohi         | 2414           | 2446 | 2444 | 3.3    | 3.2 | 3.1 |
| Mirzapur        | 2387           | 2409 | 2393 | 3.3    | 3.3 | 3.2 |
| Sonbhadra       | 2333           | 2357 | 2337 | 3      | 3   | 2.9 |

**Table A3: District-wise MPCE Estimates and %Coefficient of Variation (CV) using Direct and Model-based SAE Approach for Rural Sector**

| (1)      | (2)            |      |      | (3)    |     |     |
|----------|----------------|------|------|--------|-----|-----|
| District | MPCE Estimates |      |      | %CV    |     |     |
|          | Direct         | FH   | SFH  | Direct | FH  | SFH |
| Etah     | 3859           | 3833 | 3776 | 2.8    | 2.8 | 2.7 |
| Kasganj  | 3346           | 3372 | 3366 | 3.7    | 3.6 | 3.4 |
| Shamli   | 3999           | 3976 | 4026 | 2.9    | 2.8 | 2.7 |
| Sambhal  | 3566           | 3569 | 3557 | 2.3    | 2.3 | 2.2 |
| Hapur    | 4058           | 4043 | 4065 | 2.2    | 2.2 | 2.1 |
| Amethi   | 2910           | 2916 | 2906 | 2.4    | 2.3 | 2.3 |

*Note: FH denotes Fay–Herriot (FH) model and SFH denotes Spatial FH*

**Table A4: District-wise MPCE Estimates and %Coefficient of Variation (CV) using Direct and Model-based SAE Approach for Urban Sector**

| (1)           | (2)            |      |      | (3)    |      |      |
|---------------|----------------|------|------|--------|------|------|
| District      | MPCE Estimates |      |      | %CV    |      |      |
|               | Direct         | FH   | SFH  | Direct | FH   | SFH  |
| Saharanpur    | 4985           | 4935 | 5025 | 6.1    | 6    | 5.8  |
| Muzaffarnagar | 6308           | 6081 | 6136 | 7.5    | 7.3  | 7    |
| Bijnor        | 3562           | 3581 | 3585 | 4      | 4    | 4    |
| Moradabad     | 4081           | 4096 | 4086 | 4      | 4    | 4    |
| Rampur        | 4145           | 4153 | 4144 | 5.3    | 5.2  | 5.2  |
| Amroha        | 3008           | 3052 | 3056 | 7.3    | 7    | 7    |
| Meerut        | 5485           | 5482 | 5484 | 2.5    | 2.5  | 2.4  |
| Bagpat        | 7272           | 6995 | 6999 | 5.2    | 5.2  | 5.1  |
| Ghaziabad     | 5597           | 5597 | 5631 | 2.6    | 2.6  | 2.6  |
| Gb Nagar      | 10676          | 9705 | 9477 | 5.1    | 5.1  | 5.1  |
| Bulandshahr   | 4268           | 4272 | 4287 | 2.9    | 2.9  | 2.9  |
| Aligarh       | 4226           | 4238 | 4254 | 4.1    | 4.1  | 4.1  |
| Hathras       | 4449           | 4447 | 4426 | 5.4    | 5.2  | 5.2  |
| Mathura       | 4111           | 4135 | 4125 | 7.1    | 6.9  | 6.9  |
| Agra          | 4794           | 4808 | 4809 | 3.1    | 3.1  | 3.1  |
| Firozabad     | 4175           | 4187 | 4190 | 4.2    | 4.2  | 4.2  |
| Mainpuri      | 4142           | 4161 | 4185 | 14.8   | 13.1 | 12.7 |
| Badaun        | 4171           | 4183 | 4179 | 7.8    | 7.5  | 7.4  |
| Bareilly      | 4617           | 4628 | 4624 | 3.9    | 3.9  | 3.9  |
| Pilibhit      | 3891           | 3915 | 3918 | 6.6    | 6.4  | 6.4  |
| Shahjahanpur  | 4329           | 4341 | 4279 | 13.4   | 12   | 11.9 |
| Kheri         | 4432           | 4416 | 4406 | 5.7    | 5.6  | 5.6  |
| Sitapur       | 3664           | 3698 | 3714 | 6      | 5.8  | 5.8  |
| Hardoi        | 3809           | 3864 | 3877 | 7.8    | 7.5  | 7.4  |
| Unnao         | 4583           | 4476 | 4462 | 10.5   | 10   | 9.9  |
| Lucknow       | 6697           | 6622 | 6593 | 5      | 4.9  | 4.9  |
| Rae Bareli    | 3957           | 3979 | 3981 | 5.3    | 5.2  | 5.2  |
| Farrukhabad   | 4478           | 4473 | 4458 | 6      | 5.9  | 5.9  |
| Kannauj       | 3410           | 3449 | 3456 | 7.8    | 7.5  | 7.4  |
| Etawah        | 5701           | 5440 | 5295 | 9.5    | 9.1  | 9.1  |
| Auraiya       | 3575           | 3581 | 3583 | 3      | 3    | 3    |
| Kanpur Dehat  | 3690           | 3823 | 3831 | 15.4   | 13.5 | 13.3 |
| Kanpur Nagar  | 4944           | 4934 | 4937 | 2.6    | 2.6  | 2.6  |
| Jalaun        | 4310           | 4305 | 4315 | 4.8    | 4.8  | 4.7  |
| Jhansi        | 4801           | 4644 | 4765 | 12.4   | 11.5 | 10.7 |
| Lalitpur      | 4298           | 4288 | 4300 | 5.8    | 5.7  | 5.7  |
| Hamirpur      | 5262           | 5053 | 5074 | 9      | 8.7  | 8.5  |

**Table A4: District-wise MPCE Estimates and %Coefficient of Variation (CV) using Direct and Model-based SAE Approach for Urban Sector**

| (1)             | (2)            |      |      | (3)    |      |      |
|-----------------|----------------|------|------|--------|------|------|
| District        | MPCE Estimates |      |      | %CV    |      |      |
|                 | Direct         | FH   | SFH  | Direct | FH   | SFH  |
| Mahoba          | 4871           | 4832 | 4850 | 5.4    | 5.3  | 5.2  |
| Banda           | 4400           | 4335 | 4322 | 6.8    | 6.8  | 6.8  |
| Chitrakoot      | 3887           | 3924 | 3921 | 10.7   | 10   | 9.9  |
| Fatehpur        | 3892           | 3922 | 3941 | 6.5    | 6.3  | 6.2  |
| Pratapgarh      | 3691           | 3716 | 3698 | 5.8    | 5.7  | 5.7  |
| Kaushambi       | 3533           | 3568 | 3557 | 6.6    | 6.4  | 6.4  |
| Prayagraj       | 5043           | 5048 | 5034 | 3.2    | 3.2  | 3.2  |
| Barabanki       | 5214           | 5155 | 5180 | 5.9    | 5.7  | 5.7  |
| Faizabad        | 4969           | 4827 | 4859 | 10.6   | 10   | 9.7  |
| Ambedkar Nagar  | 2652           | 2653 | 2649 | 6.1    | 6.1  | 6.1  |
| Sultanpur       | 3598           | 3636 | 3618 | 6.7    | 6.5  | 6.5  |
| Behraich        | 4788           | 4761 | 4782 | 4.7    | 4.6  | 4.6  |
| Shrawasti       |                | 4182 | 4089 |        | 6.7  | 6.7  |
| Balrampur       | 5649           | 5087 | 5211 | 16.6   | 14.5 | 13.6 |
| Gonda           | 9889           | 8133 | 7861 | 7.9    | 8    | 8    |
| Siddharth Nagar | 2988           | 3060 | 3124 | 10.1   | 9.6  | 9.3  |
| Basti           | 3342           | 3371 | 3397 | 6      | 5.9  | 5.8  |
| Sk Nagar        | 4614           | 4505 | 4417 | 14     | 12.6 | 12.5 |
| Maharajganj     | 4135           | 4170 | 4142 | 10.7   | 10   | 9.9  |
| Gorakhpur       | 5813           | 5777 | 5763 | 3.3    | 3.3  | 3.3  |
| Kushinagar      | 3534           | 3545 | 3550 | 4.2    | 4.2  | 4.1  |
| Deoria          | 3882           | 3883 | 3877 | 3.4    | 3.4  | 3.4  |
| Azamgarh        | 3131           | 3163 | 3156 | 5.3    | 5.2  | 5.2  |
| Mau             | 2920           | 2937 | 2930 | 4.7    | 4.6  | 4.6  |
| Balliya         | 2561           | 2581 | 2575 | 5.3    | 5.2  | 5.2  |
| Jaunpur         | 3402           | 3429 | 3413 | 5.2    | 5.1  | 5.1  |
| Ghazipur        | 2689           | 2771 | 2738 | 8.4    | 8    | 8.1  |
| Chandauli       | 3381           | 3401 | 3398 | 4.8    | 4.7  | 4.7  |
| Varanasi        | 4147           | 4171 | 4148 | 4.7    | 4.6  | 4.6  |
| Bhadohi         | 3462           | 3499 | 3479 | 8.5    | 8.2  | 8.2  |
| Mirzapur        | 3888           | 3909 | 3910 | 5.3    | 5.2  | 5.2  |
| Sonbhadra       | 6585           | 5524 | 5094 | 17.3   | 14.9 | 15.2 |
| Etah            | 4354           | 4354 | 4351 | 3.6    | 3.6  | 3.6  |
| Kasganj         | 4005           | 4001 | 3993 | 7      | 6.8  | 6.8  |
| Shamli          | 6308           | 6021 | 6137 | 7.5    | 7.3  | 7.1  |
| Sambhal         | 4081           | 4087 | 4074 | 4      | 4    | 4    |
| Hapur           | 5597           | 5578 | 5582 | 2.6    | 2.6  | 2.6  |

**Table A4: District-wise MPCE Estimates and %Coefficient of Variation (CV) using Direct and Model-based SAE Approach for Urban Sector**

| (1)      | (2)            |      |      | (3)    |     |     |
|----------|----------------|------|------|--------|-----|-----|
| District | MPCE Estimates |      |      | %CV    |     |     |
|          | Direct         | FH   | SFH  | Direct | FH  | SFH |
| Amethi   | 3815           | 3828 | 3826 | 4.2    | 4.1 | 4.1 |

*Note: FH denotes Fay–Herriot (FH) model and SFH denotes Spatial FH*

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